

Blockchain Reading Task Report

Decentralized Technologies:

Decentralized technologies remove the need for a single controlling authority by distributing power, data, and decision-making across many participants(nodes). They are the foundation of blockchain, cryptocurrencies, and decentralized finance (DeFi), enabling peer-to-peer synchronizations, transparency, and resilience against single points of failure.

The main idea of decentralized technologies is to maintain transparency and security.

What is Blockchain?

Blockchain is a decentralized digital ledger technology that records transactions across a network of computers. It ensures transparency, security, and immutability without relying on a central authority.

How does Blockchain work?

- 1. Transaction initiation:**A transaction basically contains the information such as sender,receiver and amount
- 2. Broadcast to the Network:**The transaction is broadcasted to all the servers in the network and the transaction is pooled until it is validated by all the nodes.

3. Block Formation: The transactions are grouped into a block. A block consists of the timestamp, transaction data, previous block hash and a nonce which is used in finding the current block hash in mining.

4. Consensus + Validation : All the nodes validate the transaction and agree on one shared truth by using a consensus mechanism.

Proof Of Work requires the miners to compete to find the nonce to a certain target.

Proof of Stake, validators are chosen to create the next block based on how much cryptocurrency they stake.

5. Adding Block to the Chain: Once validated the block is appended to the chain. The current block is linked to the chain via the previous block hash. This makes the Blockchain **tamper-proof** , because tampering the data in one block makes the next all blocks invalid, and should re-mine all the blocks which takes a lot of work and money.

Cybersecurity Threats In Decentralized Systems:

- 1. 51% Attack:** If one miner or group controls more than half the network's power , they can dominate consensus. With majority consensus, they can rewrite recent history, reversing their own transactions to double spend coins.
- 2. Sybil Attack:** A miner creates a large number of fake node identities to gain influence over the consensus ,network's peer-to-peer decisions, routing, or voting mechanisms.
- 3. Private key Theft:** A blockchain wallet consists of 2 keys ,a **Public** and a **Private key**, a private key is like a cryptographic proof or like a

sign. If someone steals it, technically the attacker has the ownership of victims crypto-assets.

4. Eclipse/Denial of Service:An Eclipse Attack is when attackers surround and isolate a blockchain node, feed it false chain data, and exploit that isolation to cause double spending or wasted mining. It's a network-level attack that targets communication rather than raw computing power.

5. Double Spending:Spending the same cryptocurrency twice by exploiting delays in block validation. Usually this is prevented by consensus validation but in the case of Eclipse or 51% attacks this is possible because these are network level attacks and these influence the consensus.

What are Smart Contracts?

Smart contracts are self-executing programs stored on a blockchain. They automatically execute agreements when conditions in the code are met. They automatically enforce rules and execute actions without the need of a central authority.

Smart contracts are immutable and transparent .Once the code is deployed it cannot be changed. So bugs in the code can be catastrophic. The code is available for everyone to see but it cannot be altered once deployed.

Smart Contract Vulnerabilities and Attacks:

1. Reentrancy:Here a contract calls another before updating its state, opening the door to recursive calls that could drain funds or disrupt the intended execution. Furthermore, smart contracts often interact with external contracts and data sources, introducing additional vulnerabilities if these interactions are not checked appropriately.

- 2. Time Dependency Vulnerability:** refers to a vulnerability in smart contracts where the contract's logic relies on the **block.timestamp** value to make critical decisions. Since miners have limited control over the block timestamp (within a reasonable range), this can be exploited to manipulate the contract's behavior, especially in scenarios involving small time windows or incentives.
- 3. Integer Overflow and Underflow:** Smart contracts generally use unsigned integers (generally 256 bit $\rightarrow$ 0 to 2^{256}) track balances or store amounts. Sometimes when these numbers are greater than the range it overflows and causes it to wrap around 0, when these numbers occur less than 0 then they wrap around maximum value in range. This can lead to bypassing of limits and draining funds or allowing the attackers to get a lot of tokens.
- 4. Front Running:** type of attack where someone takes advantage of knowing about a pending transaction before it is confirmed, and then places their own transaction first to gain an unfair advantage.
- 5. Oracle Manipulation Vulnerabilities:** Arise when smart contracts depend on external data sources (oracles) for prices or events, and attackers influence or corrupt that external data to make it financially beneficial contract behavior.

DeFi(Decentralized Finance) Platforms:

Defi Platforms are like any other traditional finance services like banks but these are blockchain based decentralized platforms and do not depend on a central authority. So Defi Platforms basically use smart contracts to enforce rules automatically. Since Defi Platforms use smart contracts, the vulnerabilities in smart contracts also make DeFi Platforms vulnerable for ex.

Reentrancy, Front Running and Oracle manipulation. But there are some **security risks and vulnerabilities in DeFi platforms like,**

Unsecured Flash Loans: The attacker uses the flash loan without any collateral to temporarily manipulate prices, tricks the lending protocol into over-valuing their collateral, borrows real valued assets, repays the flash loan, and keeps the borrowed tokens as profit.

Rug Pulls: In decentralized finance, rug pulls commonly happen when developers launch a token and create a liquidity pool, but retain control over the liquidity or token supply once enough people invest and the tokens have a decent price, the creators remove the liquidity or dump massive token holdings, causing the token price to collapse almost instantly, and lender is left with worthless collateral.

Governance Misuse: Governance is the decision making system for a Defi system. The governance tokens gives its holders voting rights. Token holders can propose and vote on changes to the platform.

An attacker with sufficient governance tokens can submit a complex upgrade proposal with hidden malicious logic buried in technical contract code most voters won't notice hence making it profitable for the attacker.

Borrowing large numbers of governance tokens as flash loans and passing any proposal that is beneficial to the attacker before the transaction completes and the loan is repaid can often result in draining the funds.

Cross Chain Bridge Vulnerabilities:

Cross-Chain Bridges are systems that allow transfer of data between two blockchain systems. These have become one of the biggest targets to attackers in blockchain systems because these hold huge locked funds and rely on complex logic.

One of the biggest vulnerabilities in Cross-Chain Bridges are smart contract flaws. If contract logic is flawed attackers get a chance to withdraw locked funds without depositing or even drain funds.

Some real life incidents,

1. **Wormhole Incident:** An attacker exploited a flaw in Solana's signature verification, forging a fake guardian approval to mint 120,000 wETH without locking any ETH as collateral. **–Feb2022**
2. **Ronin Network Incident:** A hacker stole the private keys to make fraudulent withdrawal of assets within two transactions and the platform was defrauded for over 620 million USD. **–March2022**
3. **Nomad Bridge Incident:** The hackers exploited a faulty section of the smart contract to withdraw more assets than those deposited as collateral on the platform. attackers drained approximately 190 million USD . **–August2022**
4. **BSC Bridge Incident:** The attacker noticed a flaw in the verification of proofs by the smart contract. So the attacker tricked the system into believing a fake transaction into a legitimate one and withdrew 2 million BNB assets, the network lost 568 million USD. **–October2022**
5. **Horizon Bridge Incident:** The attackers hacked private keys from over 5000 users wallets to create fraudulent withdrawals in different tokens, hackers stole more than 100 million USD worth of ETH, BTC, and BNB. **–June2022**

Improving User-Security In Blockchain Systems:

1. **Multi-Signature Wallets:** The main idea of Multi Signature Wallets is, instead of relying on a single private, which can be susceptible to be hacked or stolen or phishing, they require multiple authorizations before a transaction can be executed.

How does it ensure security?

- Even if one key gets stolen, the funds still remain safe.
 - The attacker needs to have a certain number of keys to execute a transaction, so even if one key is bound to be stolen or hacked, the attacker cannot access the funds.
2. **Auditing and Monitoring:** Continuous auditing and monitoring of blockchain systems and smart contracts allow for the detection of suspicious activities or potential security breaches.
 3. **Being aware of phishing:** Hackers usually trick people into entering their seed phrase through suspicious emails and URLs on websites. So users should not trust such fake attempts of hackers.
 4. **Avoid using Public Wi-Fi:** Public networks are not to be trusted and are vulnerable to router interceptions, it is highly recommended to use a VPN outdoors.
 5. **Securing Devices:** Using secure devices is important because hackers can use malware to steal information from clipboard and log keystrokes.

Enhancements in WEB3.0 security:

AI in Decentralised Systems for Security:

AI models can be used to monitor blockchain transactions and detect anomalies and suspicious activities. Anomaly detection using AI prevents exploits by catching the early. Machine learning models trained on past

exploits and incidents can be used to read the smart contract code and check for even the smallest vulnerability that can be exploited. Introducing AI into DeFi Governance systems to analyze proposals for possible exploitation and vulnerabilities.

Zero-Knowledge Proofs for User Privacy:

Zero Knowledge Proofs are cryptographic protocols that let someone prove a statement is true without actually revealing the data in it. This ensures that no private data is disclosed. They allow blockchain networks to remain transparent while protecting sensitive user data.

De-centralized Identities:

A decentralized identity (DID) is a cryptographically verifiable identifier stored on a blockchain or decentralized network. Blockchain acts as a register to your credentials rather than a company's server storing your information. You don't need to completely share your information to be digitally verified, you prove the credential is valid without exposing all your personal details(**Zero Knowledge Proofs**).

Cross-Chain Interoperability:

Cross Chain Bridges are specific implementations that let you transfer tokens or data between two chains. But these are highly vulnerable to cybersecurity attacks. Cross-chain interoperability allows assets and data to move freely across blockchains. Interoperability protocols give better security than bridges because they rely on blockchain consensus instead of trusted middlemen, and use cryptographic proofs to validate cross-chain actions.

